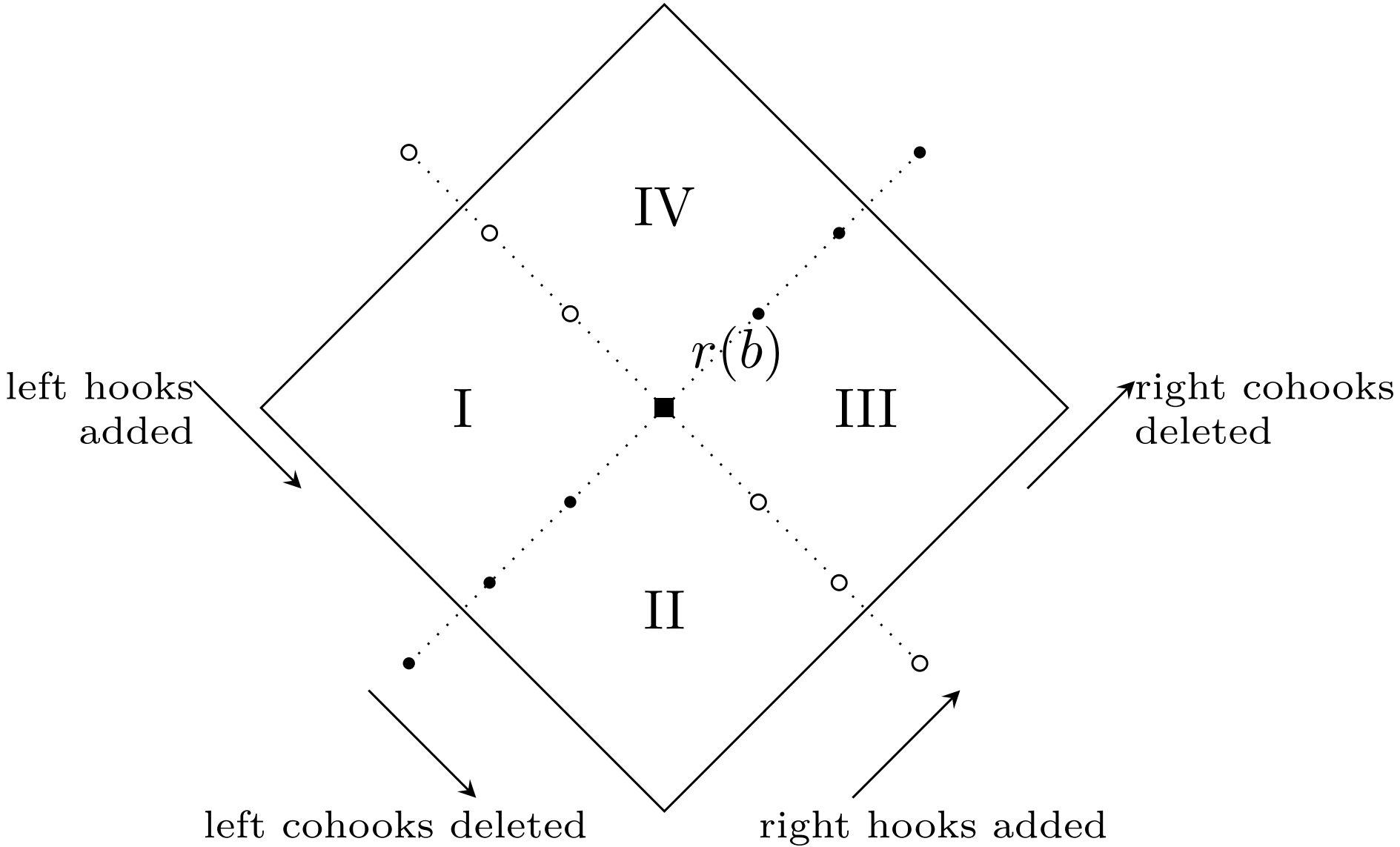
IV
I
III
II
$r(b)$
left hooks added
right cohooks deleted
left cohooks deleted
right hooks added